

ML-KEM-Based Quantum-Resilient Checkpoint File Transfer Protocol for 6G Networks

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Abstract—Millimetre-wave 5G access already suffers frequent link disruption from blockage and dense small-cell handover; beyond-5G (B5G) deployments intensify this, and 6G pushes into the terahertz band, where the same physics is more severe still. Secure bulk transfer is doubly penalised: the session must be re-established after every outage, and the public-key exchange protecting it is the component most exposed to harvest-now, decrypt-later quantum cryptanalysis. This paper presents a quantum-resilient checkpointed file transfer protocol that establishes session keys with ML-KEM-768 (NIST FIPS 203), protects bulk data with AES-256-GCM under an index-derived deterministic nonce, and persists a receiver-authoritative, MAC-protected checkpoint so that an interrupted transfer resumes from the last verified chunk without repeating the key exchange or re-sending verified data. A closed-form delay model, parameterised by link rate and disruption rate, spans the 5G, B5G and 6G regimes and yields the checkpoint interval minimising total overhead. Six performance metrics are evaluated at one operating point per generation: at 6G scale (100 MB file, 1 Gbps link) the model projects a 24.1% shorter completion time and a 23.8× reduction in redundant re-sent data for a transfer interrupted five times, rising to 29.3% at B5G and 36.1% at 5G rates, with transfer efficiency improving from 28.6% to 90.5% in every case.

Keywords—post-quantum cryptography, ML-KEM, FIPS 203, 6G, terahertz, checkpoint/resume, secure file transfer, AES-256-GCM, performance metrics.

I INTRODUCTION

Link disruption is not a future problem. 5G mmWave access already suffers frequent outage from hand and body blockage, and fast conditional handover mechanisms exist precisely because dense small-cell mobility interrupts sessions routinely [8]. B5G deployments push this further with ultra-dense and non-terrestrial links [9], and 6G sharpens the same problem at THz frequencies, where severe path and penetration loss let a hand, a wall or a passing vehicle sever a beam [7]. Bulk transfers are therefore interrupted many times before they complete, and the severity grows along that progression.

Conventional key agreement (RSA, ECDH) falls to Shor’s algorithm, and traffic recorded today can be decrypted retroactively once a cryptographically relevant quantum computer exists [2]. NIST has standardised the module-lattice KEM ML-KEM in FIPS 203 [1]; its public key and ciphertext are an order of magnitude larger than elliptic-curve equivalents [3, 4]. A protocol that restarts after each outage pays that enlarged handshake repeatedly. Resumable mechanisms such as FTP restart [12] assume an unencrypted or per-connection-keyed channel, while post-quantum evaluations for mobile networks measure handshake latency, energy or authentication cost on clean, uninterrupted links [6, 5, 13, 14] and do not consider what happens when the transfer itself is interrupted mid-stream. This work occupies that gap, and is parameterised (Section II) so that one model and one protocol cover 5G, B5G and 6G rather than a single generation.

A Related Work

Recent work on post-quantum wireless security falls into three groups. The first measures handshake cost in isolation: Wirth *et al.* [3] and Zheng *et al.* [4] benchmark ML-KEM inside TLS 1.3 and find the overhead small once amortised, but assume a channel that does not drop. The second targets mobile-network authentication rather than bulk data: Vuppala *et al.* [13] propose hybrid primary authentication for 6G UE onboarding, and Khan *et al.* [5] analyse PQ key exchange for 5G networking; both stop at session establishment. The third measures PQC

cost on constrained endpoints [14, 6] — again a single, uninterrupted handshake. The mobility literature treats disruption itself as the object of study [8, 9] but carries no post-quantum layer. None combine a post-quantum handshake with a resumable data plane, which is where this contribution sits.

B Contributions

1. A protocol combining ML-KEM-768 key establishment, AES-256-GCM bulk protection and checkpoint/resume, parameterised to apply across 5G, B5G and 6G link conditions.
2. A safety argument for resume: a deterministic, chunk-index-derived GCM nonce plus a receiver-authoritative HMAC-protected checkpoint that cannot be advanced or rewound by a forged offset.
3. A closed-form delay model and the checkpoint interval N^* minimising persistence cost plus outage rework.
4. A six-metric performance evaluation at one operating point for each of 5G, B5G and 6G.

II SYSTEM MODEL

A Network and Disruption Model

Sender and receiver are endpoints (UEs, or a UE and an edge server) whose path traverses the air interface of the deployment in question. The link is modelled generically as a lossy pipe of capacity R b/s with round-trip time τ and an outage process of rate λ ; a transfer of duration T sees $n_d = \lambda T$ expected disruptions, each costing a detection delay T_d and a reconnection delay T_r . The model is generation-agnostic: R, τ, T_d, T_r and λ differ per generation, but the protocol does not. Table 1 fixes one operating point for each of 5G mmWave, B5G and 6G THz; λ rises and τ falls with carrier frequency and cell density, while R rises towards the tens of Gbps projected for 6G [7].

B Cryptographic Model

A file of S bits is split into $n_c = \lceil S/c \rceil$ chunks of c bits. The session begins with ML-KEM-768 encapsulation ($|pk| = 1184$ B, $|ct| = 1088$ B, shared secret 32 B), costing

$$T_h = t_{kg} + t_{enc} + t_{dec} + \frac{|pk| + |ct|}{R} + \tau, \quad (1)$$

which is RTT-dominated, since the lattice operations are microsecond-scale. Keys are derived once per session as

$$K \parallel K_{ck} = \text{HKDF-SHA256}(ss, sid, \text{"pq6g/v1"}), \quad (2)$$

where K is the AEAD key and K_{ck} keys the checkpoint MAC [10]. Chunk i is sealed as

$$C_i = \text{AES-256-GCM}_K(\nu_i, M_i, A_i), \quad \nu_i = sid_{32} \parallel i_{64}, \quad (3)$$

with associated data $A_i = (sid, i, |M_i|)$ [11]. Because ν_i is a pure function of the session id and chunk index, re-sending chunk i after an outage is idempotent and cannot reuse a nonce with different plaintext — the property that makes resume compatible with GCM. A rolling digest $H_i = \text{SHA-256}(H_{i-1} \parallel \text{SHA-256}(C_i))$ tracks integrity, $H_{-1} = \emptyset$. A checkpoint is

$$\chi = (sid, i^*, o^*, H_{i^*}, \text{HMAC}_{K_{ck}}(sid \parallel i^* \parallel o^* \parallel H_{i^*})), \quad (4)$$

written durably by the receiver after verifying each chunk's tag. The sender never dictates the resume point, and the MAC binds the offset to the session key, so an off-path attacker cannot force a rewind or a silent skip.

III PROPOSED FRAMEWORK

A Architecture

Fig. 1 shows the architecture: the sender splits the input into sequenced chunks, runs ML-KEM-768 and the key schedule of (2), seals each chunk per (3), and tracks acknowledged state in the checkpoint manager; the receiver mirrors this and owns the durable checkpoint store and reassembly engine. The channel emulator is the only component that injects impairment, so all evaluation arms see statistically identical conditions and any performance gap is attributable to the protocol, not to channel variance. Sender and receiver communicate over a single custom-framed TCP connection; the wire frame is $[sid | type | seq | len | \nu_i | C_i | tag]$, with distinct types for handshake, data, acknowledgement, resume request and resume response.

B Control Flow and Algorithm

Fig. 2 traces the same logic as a state machine. The upper row is the steady transmit path: encapsulate, split, encrypt-and-send, and on each acknowledgement write a checkpoint every

Table 1: Prototype stack, fixed parameters, and the three per-generation operating points used in the evaluation.

Layer	Component
Key establishment	ML-KEM-768 (FIPS 203), <code>liboqs/kyber-py</code>
Key schedule	HKDF-SHA256 $\rightarrow$ AEAD + checkpoint MAC key
Bulk encryption	AES-256-GCM, deterministic nonce $sid \parallel i$
Transport	TCP sockets, custom framed protocol
Channel emulation	token bucket, jitter, Poisson blockage; <code>netem</code>
Inspection / eval.	Wireshark + Lua dissector; 3-arm harness
File / chunk / t_{ck}	100 MB / 64 KB / 5 ms, interval $N = 64$

Parameter	5G mmWave	Beyond-5G	6G THz
Link rate R	800 Mbps	900 Mbps	1 Gbps
Round-trip time τ	2 ms	1.5 ms	1 ms
Detect / reconnect	0.3 / 0.3 s	0.4 / 0.4 s	0.5 / 0.5 s
Blockage rate λ	0.8 s^{-1}	1.0 s^{-1}	1.2 s^{-1}

Algorithm 1 Checkpointed quantum-resilient transfer (sender)

Input: file F , receiver address, chunk size c , interval N

- 1: $(pk) \leftarrow \text{RCV}(\text{HELLO}); (ct, ss) \leftarrow \text{MLKEM.Encaps}(pk)$
- 2: $\text{SEND}(ct); K \parallel K_{ck} \leftarrow \text{HKDF}(ss, sid)$
- 3: split F into $M_0 \dots M_{n_c-1}; i \leftarrow 0; H_{-1} \leftarrow \emptyset$
- 4: **while** $i < n_c$ **do**
- 5: $\nu_i \leftarrow sid \parallel i; C_i \leftarrow \text{SEAL}_K(\nu_i, M_i, A_i); \text{SEND}(C_i)$
- 6: **if** $\text{ACK}(i)$ received **then**
- 7: $H_i \leftarrow H(H_{i-1} \parallel H(C_i))$
- 8: **if** $i \bmod N = 0$ **then**
- 9: $\text{PERSIST}(\chi)$
- 10: $i \leftarrow i + 1$
- 11: **else if** disruption detected **then**
- 12: PAUSE; reconnect with exponential backoff
- 13: $\chi \leftarrow \text{REQUESTRESUME}(sid)$
- 14: abort if $\text{HMAC}_{K_{ck}}(\chi)$ invalid or H_{i^*} mismatches
- 15: $i \leftarrow i^* + 1$
- 16: $\text{SEND}(\text{FIN}, \text{SHA-256}(F));$ verify receiver digest

N chunks. A disruption diverts to the lower row, which pauses, backs off exponentially and issues a single-RTT RESUME request carrying only the session id. The lower row re-joins the upper row at exactly the point it left, so recovery never revisits the handshake box. Algorithm 1 states the sender loop: recovery re-enters the transmit loop directly, with no key exchange, no re-derivation and no retransmission below index i^* .

C Optimal Checkpoint Interval

With n_d expected disruptions, the total overhead of persistence plus expected rework is

$$O(N) = \underbrace{\frac{S}{Nc} t_{ck}}_{\text{persistence}} + \underbrace{\frac{n_d Nc}{2R}}_{\text{expected rework}}, \quad (5)$$

convex in N ; setting $dO/dN = 0$ gives

$$N^* = \sqrt{\frac{2RS t_{ck}}{n_d c^2}}. \quad (6)$$

For $S = 100$ MB, $c = 64$ KB, $R = 1$ Gbps, $t_{ck} = 5$ ms and $n_d = 5$: $N^* \approx 76$ chunks and $O(N^*) \approx 0.2$ s. The interval should widen on fast links and narrow as blockage rises.

IV IMPLEMENTATION, RESULTS AND PERFORMANCE ANALYSIS

The protocol is being implemented in Python 3.10 over TCP sockets with a Wireshark dissector and a three-arm benchmark harness (*classic-restart*, *ML-KEM-restart*, *ML-KEM-checkpoint*). **The results below are analytical projections from (1)–(6), not measured trials;** empirical validation follows once the socket-level prototype is complete.

A Performance Metrics

Six metrics are used, each comparable across generations: (i) *completion time* $T(n_d)$, the wall-clock time to deliver and verify the file under n_d disruptions; (ii) *effective goodput* $G = S/T(n_d)$, which folds outage cost into a single rate; (iii) *transfer efficiency* $\eta = S/(S + S_{\text{red}})$, isolating wasted airtime and energy from raw speed; (iv) *per-outage recovery cost* $T_{\text{rec}} = T_d + T_r + \tau + Nc/2R$ for the proposal against $T_d + T_r + T_h + \epsilon + \bar{p}S/R$

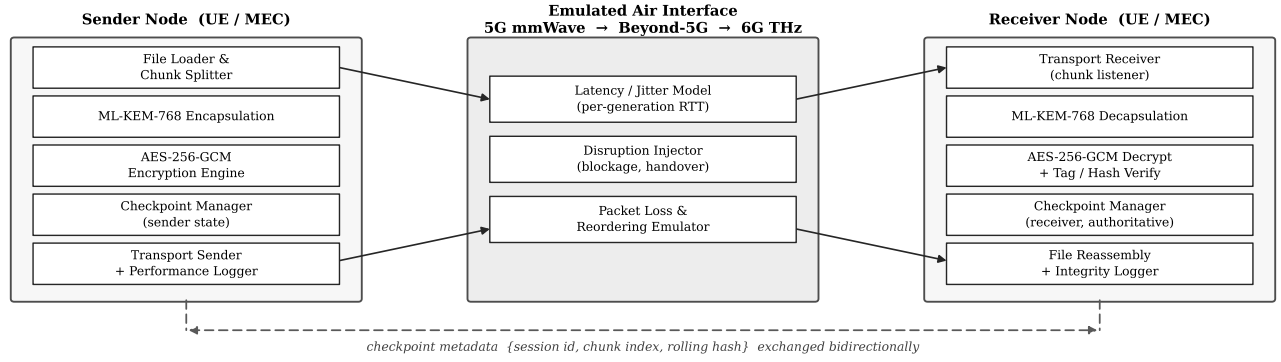


Figure 1: System architecture. Sender and receiver run symmetric crypto, checkpoint and transport stacks either side of the emulated air interface — the sole source of latency, jitter, blockage and handover impairment, instantiated in turn at the three operating points of Table 1.

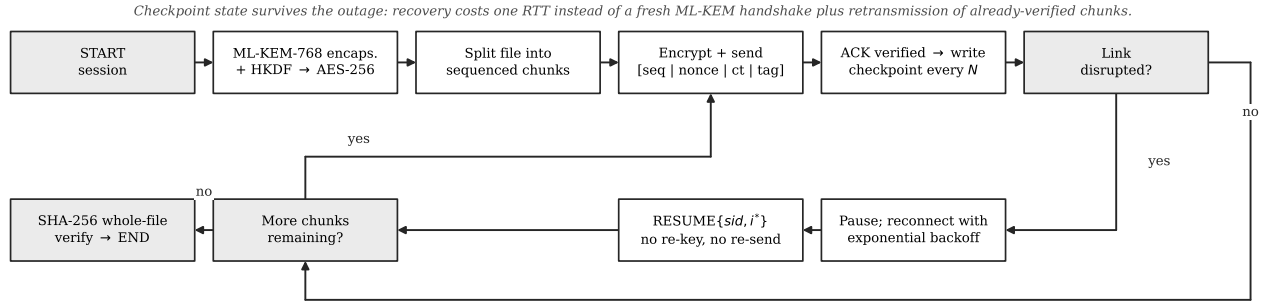


Figure 2: Control flow. The recovery path (lower row) re-enters the transmit loop after a one-RTT RESUME exchange; the ML-KEM handshake and every chunk up to i^* are never repeated. The flow is identical for all three generations.

for the baseline, where $\bar{p}$ is the mean fraction already sent; (v) *handshake amortisation* $\alpha = n_d + 1$, the number of ML-KEM-768 exchanges the baseline pays for the proposal's one; and (vi) *checkpoint overhead ratio* $\rho = (S/Nc)t_{ck}/(S/R)$, the price of the mechanism on an undisrupted link.

B Results

At 6G scale, completion time falls by 24.1% at five disruptions and 26.2% at ten, Fig. 3(a). Redundant data, Fig. 3(b), falls from 250 MB to 10.5 MB at five disruptions — a $23.8\times$ reduction — and the gap widens with the interruption point, since a later disruption leaves more verified data for the baseline to discard. On an undisrupted transfer the checkpoint writes cost 0.12 s, making the proposal 14.9% slower ($\rho = 0.149$); the crossover is at the first disruption, so the protocol wins on any link disrupted at least once.

Applied along the generation progression, the same closed form gives the monotone picture of Table 2 and Fig. 4. The relative gain is *largest at 5G* and shrinks towards 6G — 36.1%, 29.3% and 24.1% at five disruptions — because the fixed per-outage detection and reconnection cost dominates more heavily at the lower link rate. Goodput rises from 15.1 to 23.6 MB/s at 5G, 13.8 to 19.6 MB/s at B5G and 12.6 to 16.6 MB/s at 6G, while per-outage recovery cost falls by 44.6%, 35.3% and 28.5%. Transfer efficiency improves identically in all three cases, from 28.6% to 90.5%, since η depends only on file size and checkpoint interval. Handshake amortisation is likewise rate-invariant: the baseline sends 13 632 B of key material over six exchanges where the proposal sends 2272 B once. AEAD

Table 2: Projected performance metrics at five disruptions. B = baseline restart, P = proposed checkpoint.

Metric	5G mmWave		Beyond-5G		6G THz	
	B	P	B	P	B	P
Completion time (s)	6.63	4.24	7.23	5.11	7.92	6.01
reduction		36.1%		29.3%		24.1%
Goodput (MB/s)	15.1	23.6	13.8	19.6	12.6	16.6
Redundant data (MB)	250	10.5	250	10.5	250	10.5
Transfer efficiency	28.6%	90.5%	28.6%	90.5%	28.6%	90.5%
Recovery / outage (s)	1.13	0.62	1.27	0.82	1.42	1.02
Handshake material (B)	13 632	2272	13 632	2272	13 632	2272
Checkpoint overhead ρ	—	11.9%	—	13.4%	—	14.9%

expansion is a negligible 16 B tag per 64 KB chunk.

Fig. 3(c) shows the interval trade-off: flat near the optimum, f_{sync} -dominated below $N \sim 10$ and rework-dominated above $N \sim 500$. Fig. 4(c) makes the crossover explicit — all three curves start negative at $n_d = 0$ and turn positive at the first disruption.

C Discussion

ML-KEM-768 is computationally cheaper than P-256 ECDH but its pk/ct are more than $17\times$ larger — negligible once per session, punitive once per outage. Checkpointing therefore complements a post-quantum handshake rather than being an independent optimisation: the two techniques target exactly the two costs a disrupted, post-quantum link incurs. The metrics separate into those that improve most where the link is slowest (completion time, goodput, recovery cost) and those that are rate-invariant (transfer efficiency, handshake amortisation), so

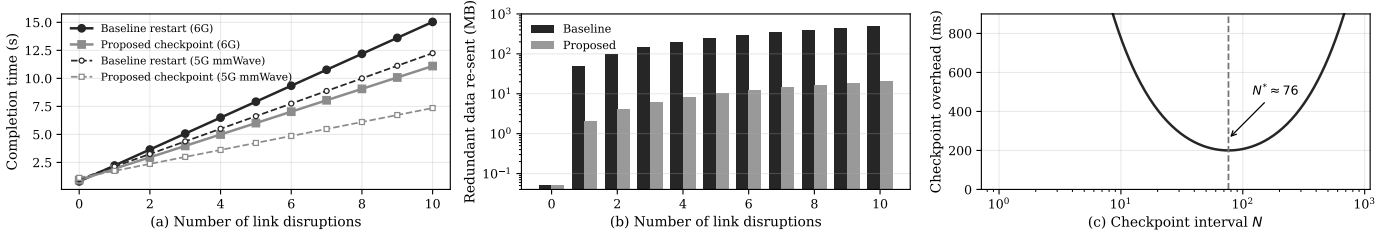


Figure 3: Analytical projection ($S = 100$ MB, $N = 64$): (a) completion time at the 6G THz and 5G mmWave points; (b) redundant data re-sent, log scale; (c) checkpoint overhead versus N , minimised at $N^* \approx 76$.

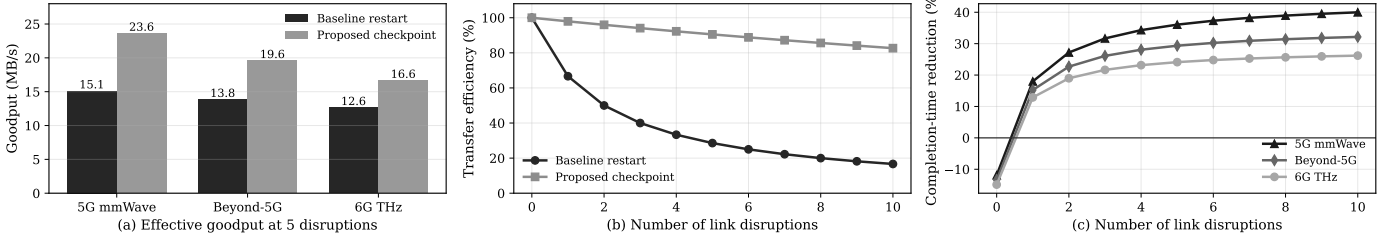


Figure 4: Performance metrics across the 5G → B5G → 6G progression: (a) effective goodput at five disruptions; (b) transfer efficiency, rate-invariant and hence common to all three; (c) completion-time reduction, negative only on an undisrupted link.

the benefit is already available at today’s 5G rates. The handshake itself is an unauthenticated KEM exchange, resisting a passive eavesdropper but not an active machine-in-the-middle; the checkpoint MAC protects the resume mechanism against a forged offset, not the peer’s identity, which is why endpoint authentication (ML-DSA) is immediate future work.

V CONCLUSION AND FUTURE WORK

We presented a checkpointed, quantum-resilient file transfer protocol for disruption-prone links, combining ML-KEM-768 key establishment, AES-256-GCM under an index-derived deterministic nonce, and a receiver-authoritative MAC-protected checkpoint that makes resume both safe and cheap. Across the six metrics the model indicates that the advantage grows precisely where post-quantum key material is most expensive — links that break often and late in a transfer — at a cost of 11.9–14.9% overhead on uninterrupted transfers, recovered at the first outage. The gain holds at 5G, B5G and 6G parameters alike and is proportionally largest at the lower link rate. The next step is the socket-level prototype and the empirical three-arm benchmark validating these projections. Subsequent work will add ML-DSA (FIPS 204) endpoint authentication, tune N adaptively against the observed blockage rate per (6), host checkpoint state at a MEC node, and replace the statistical channel model with an ns-3 propagation model.

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